

Understanding Blockchain Technology

: How It Works and Why It's Popular

A. Warming Up

- Have you heard of blockchain technology before? If so, what do you understand about it?
 무엇이냐 다 개념 이해를 여러 권록 되어 명상 저장하여
 기록을 해가 구조적으로 접근하는 것을
- Can you think of any industries or sectors where blockchain technology could have a significant impact?
 은행, 금융, 부동산, 의료, 제조업, 유통업, 에너지, 공공 서비스
 거래에 대한 기록을 안전하게 공유하는
- What are some of the reasons why blockchain technology has gained popularity in recent years?
 투명, 분산성, 보안성, 거래의 투명성, 거래 기록
 신뢰성을 제공

B. Key Expressions

- Over the past few years, the term "blockchain technology" has become increasingly familiar, particularly in relation to cryptocurrencies like Bitcoin.

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- What role does the peer-to-peer network play?
 (A) It ensures the accuracy and integrity of the shared ledger
 (B) It stores and processes transactions
 (C) It controls and regulates the supply of cryptocurrencies
 (D) It facilitates communication between central authorities and users
- What does blockchain technology eliminate the need for in transactions?
 (A) Digital signatures
 (B) Intermediaries
 (C) Cryptographic keys
 (D) Computing power
- What advantage does blockchain technology offer over traditional banking systems?
 (A) Lower transaction costs
 (B) Faster transaction processing
 (C) Enhanced security and transparency
 (D) Increased privacy for users

E. Grammar Checkup

Choose one that can best complete the following sentences.

- Over the past few years, the term "blockchain technology" _____ increasingly familiar, particularly in relation to cryptocurrencies like Bitcoin.

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D. Comprehension Checkup

Choose the best answer to the following questions.

- What is blockchain technology?
 (A) A method of recording information that is easy to change and manipulate
 (B) A centralized ledger controlled by governments or banks
 (C) A distributed ledger that makes it difficult to change or hack information
 (D) A technology used exclusively for cryptocurrencies
- What is a decentralized system in the context of blockchain technology?
 (A) Transactions controlled by central authorities like governments or banks
 (B) Transactions conducted through mutual consensus among users
 (C) Transactions requiring intermediaries for approval
 (D) Transactions that involve multiple parties with shared control
- What is the purpose of cryptographic keys in blockchain technology?
 (A) To establish a unique digital identity for each user
 (B) To facilitate communication between central authorities and users
 (C) To increase the speed of transaction processing
 (D) To control the supply of cryptocurrencies

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- Many people find _____ challenging to grasp the true meaning and significance of blockchain.
 (A) it (B) to become
 (C) becoming (D) has become
- This enhanced security provides confidence in the integrity of transactions and protects _____ unauthorized access.
 (A) in (B) to
 (C) for (D) against
- Unlike traditional transactions that require approval from central authorities like governments or banks, blockchain transactions _____ conducted through a mutual consensus among users.
 (A) is (B) are
 (C) to be (D) being
- Each individual has their own pair of keys, _____ they use to establish a unique digital identity.
 (A) who (B) which
 (C) what (D) that
- Blockchain technology offers a secure, efficient, and decentralized approach to _____ and verifying transactions.
 (A) record (B) to record
 (C) recording (D) recorded

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F. Vocabulary Checkup

Choose one that can best complete the following sentences.

- Blockchain technology is a method of recording information in a way that makes it nearly impossible to change, hack, or _____.
(A) maintain (B) manipulate
(C) record (D) introduce
- At its core, blockchain is a distributed ledger that _____ and distributes transactions across a network of computers.
(A) prevents (B) replicates
(C) understands (D) interferes
- The popularity of blockchain technology stems from its ability to address _____ inherent in traditional transactions.
(A) personalities (B) abilities
(C) mechanisms (D) vulnerabilities
- Recognizing this flaw, third-party payment applications have _____ in recent years.
(A) included (B) modified
(C) conducted (D) emerged
- These participants _____ digital signatures to achieve consensus on transactions and other matters.

(A) blame

(C) utilize

(B) record

(D) praise

- By eliminating intermediaries and enhancing security, blockchain has the potential to revolutionize various industries, _____ faster, more transparent, and cost-effective transactions.

(A) agreeing with
(C) originating from

(B) interfering with
(D) leading to

G. Guided Composition

Using the useful expressions given, put the following Korean into English.

- 지난 몇 년 동안 "블록체인 기술"이라는 용어는 점점 더 친숙해졌다.
(over the past few years)
Over the past few years, the term "blockchain technology" has become increasingly familiar.
- 블록체인 기술은 변경, 해킹, 조작이 거의 불가능한 방식으로 정보를 기록하는 방법이다.
(makes it impossible to)
Blockchain technology is a method of recording information in a way that makes it nearly impossible to change, hack, or manipulate.

Science & Technology Reading

- 블록체인 기술의 인기는 기존 거래에 내재된 취약점을 해결할 수 있는 능력에서 비롯된다.
(stem from)
The popularity of blockchain technology stems from its ability to address vulnerabilities inherent in traditional transactions.
- 암호화 키는 개인 키와 공개 키로 구성된다.
(consist of)
Cryptographic keys consist of a private key and a public key.
- 이 디지털 식별은 일반적으로 '디지털 서명'이라고 불린다.
(be referred to as)
This digital identity is commonly referred to as a "digital signature".